

JavaScript Assignment: Operators and Decision Making

Instructions

Answer all questions carefully. Write clean and understandable code where required.

Use of AI and copying code are strictly prohibited.

Section A: Theory Questions

1. Define the following JavaScript operators and give one example each:

- a. Arithmetic Operator
 - b. Comparison Operator
 - c. Logical Operator
 - d. Assignment Operator
 - e. Ternary Operator
-

2. Explain the difference between:

- a. == and ===
 - b. && and ||
 - c. if...else and switch
-

3. What will be the output of the following code?

```
let a = 10;
```

```
let b = "10";
```

```
console.log(a == b);
```

```
console.log(a === b);
```

```
console.log(a > 5 && b < 20);
```

Explain your answer.

Section B: Practical Questions

4. Write a JavaScript program that checks whether a number is even or odd.

Example:

- Input: 7
 - Output: 7 is Odd
-

5. Write a JavaScript program that checks whether a person is eligible to vote.

Condition:

- If age is 18 and above → "Eligible to vote"
 - Otherwise → "Not eligible to vote"
-

6. Write a JavaScript program using if...else if...else to grade a student based on score:

- 70 – 100 → A
 - 60 – 69 → B
 - 50 – 59 → C
 - 40 – 49 → D
 - Below 40 → F
-

7. Rewrite Question 6 using switch statement where applicable.

8. Write a JavaScript program using the ternary operator to check if a number is positive or negative.

Example:

let num = -5;

Expected Output:

Negative Number

Section C: Challenge Question

9. Build a simple calculator using JavaScript operators.

The program should accept:

- First number
- Second number

- Operator (+, -, *, /)

Then display the correct result using decision statements.

Example:

Input:

num1 = 10

num2 = 5

operator = "*"

Output:

Result = 50

Bonus Question

10. Write a JavaScript program that determines the largest among three numbers.

Example:

Input: 12, 45, 8

Output: 45 is the largest number

Submission Note

Submit your assignment with proper indentation and comments in your code.

Good luck.